

Chase M. Enlowsmith

CONTACT INFORMATION

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EDUCATION

The University of Texas at Austin, Austin, Texas Dec 2025
Bachelor of Science in Physics, Bachelor of Science in Astronomy
• GPA 3.91
• Relevant Coursework: Quantum Mechanics, Intro to Quantum Field Theory, Electrodynamics, Thermodynamics, Relativity, Astrophysics, Cosmology

HONORS AND AWARDS

Phi Beta Kappa, Alpha of Texas chapter Feb 2025 – Current
University Honors, University of Texas at Austin May 2023 – Dec 2025
Lightning Talk Competition - 2nd Place, Applied Research Labs Jul 2025
2025 Honors Scholar Lightning Talk:
“Machine Learning on Sound Speed Profiles to Characterize Ocean Variability”

PROFESSIONAL RESEARCH EXPERIENCE

Applied Research Lab, Austin, Texas Jun 2025 – Jun 2026
Honors Scholar Intern
• Engineered an unsupervised machine learning pipeline using Gaussian Mixture Models (GMM) to geographically cluster multidimensional oceanographic profiles across basin scales and successfully characterize ocean variability.
• Developed a bespoke, physically-motivated data-reduction algorithm to robustly handle sparse datasets; utilized cross-validation to iteratively inform the decomposition, successfully avoiding nonphysical data extrapolation and pushing pipeline accuracy and efficiency beyond industry standards.
• Programmed an interactive graphical user interface (GUI) to maximize model interpretability and aid in decision making, enabling users to dynamically visualize cluster boundaries, evaluate GMM confidence metrics, and analyze geographic regions across various parameter spaces and slices.
• Collaborated with an interdisciplinary team to translate data-driven insights into actionable strategies for tactical oceanographic decision-making.
• Authoring a manuscript to be published in peer-reviewed journals.

ACADEMIC RESEARCH EXPERIENCE

Center for Gravitational Physics, U. Texas, Austin Feb 2026 – Present
Supervisor: Digvijay Wadekar
• Built random-forest and symbolic-regression pipelines to derive closed-form analytic proxies for galaxy-cluster mass and the hydrostatic mass bias from observable galaxy features, using MillenniumTNG cosmological simulations.
• Reproduced and validated an analytic correction to the Sunyaev–Zel’dovich flux–mass relation, reducing cross-validated prediction scatter by 20–30%.
• Designed an autonomous LLM-agentic discovery workflow with a leakage-resistant scoring harness to propose and evaluate physics-informed closed-form analytic proxies of galaxy cluster mass.
• Applied for NeurIPS 2026 satellite workshop and authoring a first-author manuscript to be published in peer-reviewed journals.

Center for Gravitational Physics, U. Texas, Austin

Oct 2024 – Aug 2025

Supervisor: Hsin-Yu Chen

- Prototyped Redrock-based redshift fitting for one-dimensional spectra from the Hobby-Eberly Telescope VIRUS Parallel Survey (HETVIPS), testing a subset of spectra and identifying computational bottlenecks in adapting the software to the survey data.
- Investigated Bayesian dark-siren inference combining gravitational-wave distance information with galaxy redshifts, probing how the number of lines of sight and galaxy-catalog sparsity shape the posterior for the Hubble constant, H_0 .
- Cross-validated an in-house pipeline against a published code, resolved a discrepancy in their posteriors, and characterized how the posterior broadens with catalog size across the 497 lines of sight of a real O4 skymap (S250207bg).

RESEARCH
PROJECTS

CMBverse, U. Texas, Austin

Jun 2024 – Present

- Built and deployed *CMBverse*, an interactive web tool for academics to explore CMB power spectra across a 6-parameter Λ CDM space in real time (TensorFlow.js, WASM backend).
- Developed the underlying end-to-end neural emulator of the CLASS Einstein Boltzmann solver (built on CONNECT/TensorFlow), achieving sub-percent accuracy over cosmologically relevant parameter space while accelerating spectrum evaluation by $\sim 4 \times 10^4$ over direct Boltzmann integration.
- Authoring a first-author manuscript to be published in peer-reviewed journals.

Gravitational Lens Detection, U. Texas, Austin

Sep 2025 – Dec 2025

- Engineered an unsupervised machine learning pipeline utilizing t-Distributed Stochastic Neighbor Embedding (t-SNE) to accurately identify gravitationally-lensed galaxies within a sample of luminous red galaxies in the Kilo Degree Survey (KiDS) DR5.
- Designed a theoretical extension to the pipeline, evaluating how decomposing galaxy imaging tiles into a statistical latent space via a Variational Autoencoder (VAE) would optimize the detection success-rate of lens candidates.

GW Data Sonification, U. Texas, Austin

Sep 2025 – Dec 2025

- Developed an open-source command-line interface (CLI) to download, visualize, and sonify gravitational wave events from the GWOSC (Gravitational Wave Open Science Center), translating physical strain data into accessible audio formats.
- Authored an interactive documentation via Jupyter Notebook as a guide to the software usage and underlying physics driving the signal generation.

LEADERSHIP AND
COMMUNITY
INVOLVEMENT

Theta Chi Fraternity, U. Texas, Austin

Scholarship Chair

Dec 2023 – Dec 2024

- Connected 40+ members with applicable scholarship opportunities, career events, and academic opportunities.
- Oversaw academic performance and provided one-on-one career and academic assistance.

Philanthropy Committee

Jan 2025 – Jun 2025

- Collaborated with a committee to plan and lead a series of on and off-campus philanthropy events to fundraise for the United Services Organization (USO).

UT MicroFarm, Austin, Texas

Apr 2023 – May 2024

Volunteer

- Worked with a team of volunteers to grow plants and manage the upkeep of community garden plots.
- Contributed to producing dozens of pounds of fresh produce donated to the student food bank.

TECHNICAL SKILLS	Programming: Python, JavaScript, HTML/CSS, Bash, MATLAB, $\text{\LaTeX}$ ML & Statistics: TensorFlow/Keras, TensorFlow.js, scikit-learn, PySR; neural-network emulation, symbolic regression, Gaussian mixture models, t-SNE, Bayesian inference, LLM agents (Anthropic API, Claude Code) Scientific Computing: NumPy, SciPy, pandas, Astropy, Matplotlib, Jupyter; parallel computing (MPI) Astro & Cosmology: CLASS, CONNECT, Redrock, HEALPix/FITS; MillenniumTNG simulations, GWOSC data Tools: Git/GitHub, Unix, Microsoft Office, Google Workspace
LANGUAGES	English (Native), Spanish (Basic)